

cho

①

Numbers :

1- Decimal : base = 10 (D or d) default
0, 1, 2, ..., 9

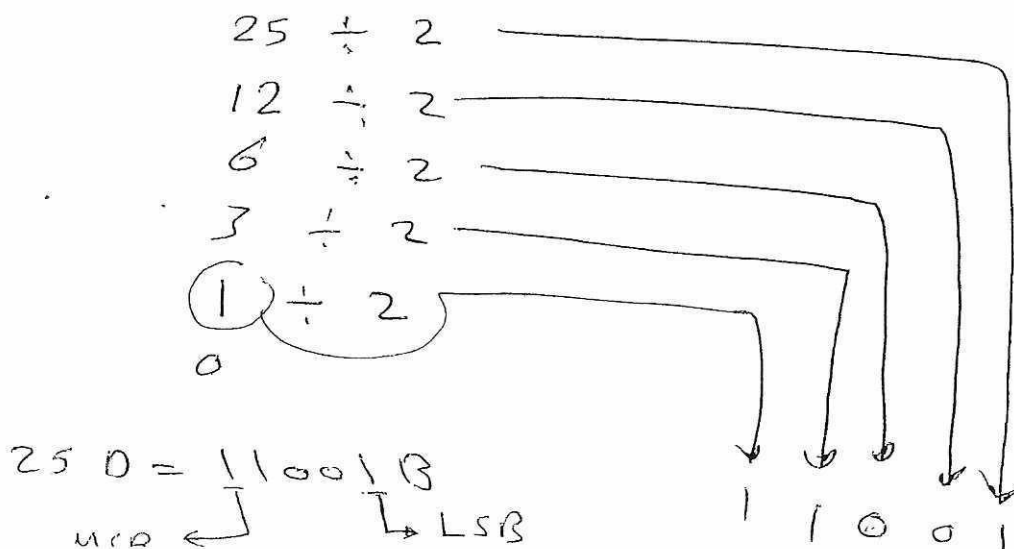
2- Binary : base = 2 (B or b)
0, 1

3- Hexadecimal : base = 16 (H or h)
0, 1, 2, ..., 9, ¹⁰A, ¹¹B, ¹²C, ¹³D, ¹⁴E, ¹⁵F

I - Convert from decimal $\rightarrow$ Any system

1- Decimal $\rightarrow$ Binary

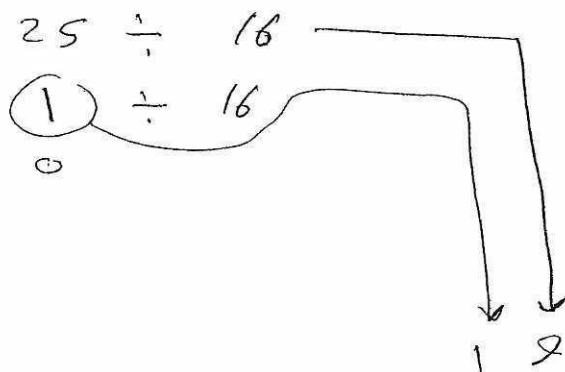
$$(25)_{10} = (\quad ? \quad)_2$$



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② Decimal $\rightarrow$ Hex

$$(25)_{10} = (?)_{16}$$

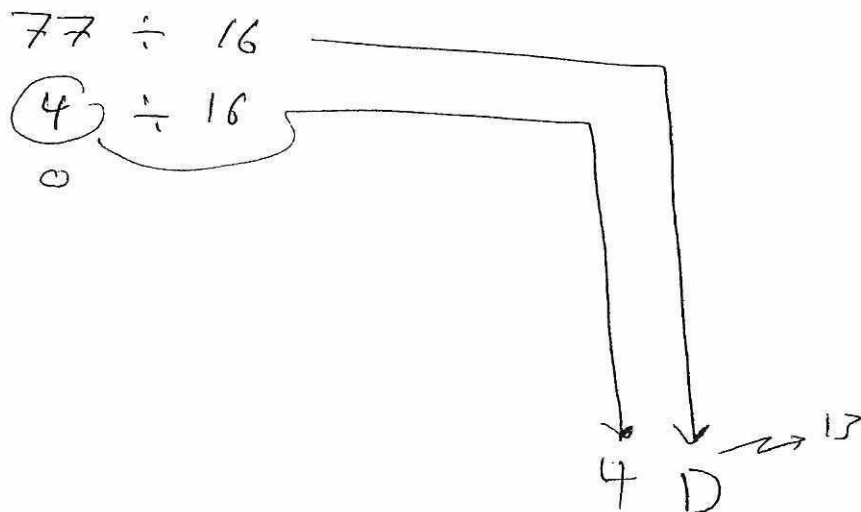


$$(25)_{10} = (19)_{16}$$

$$25D = 19H$$

$$(77)_{10} = (?)_{16}$$

A.6?



$$77D = 4DH$$

② Convert from Any system $\rightarrow$ Decimal ③

① Binary $\rightarrow$ Decimal

$$(110110)_2 = (?)_{10}$$

5 4 3 2 1 0

$$\begin{aligned} 110110 &= 0 \times 2^0 + 1 \times 2^1 + 1 \times 2^2 + 0 \times 2^3 + 1 \times 2^4 + 1 \times 2^5 \\ &= 0 + 2 + 4 + 0 + 16 + 32 \\ &= 54 \end{aligned}$$

$$(110110)_2 = 54_{10}$$

طريقة أخرى

32 16 8 4 2 1

$$\begin{aligned} 110110 &= 2 + 4 + 16 + 32 \\ &= 54 \end{aligned}$$

② Hex $\rightarrow$ Decimal

$$(2AF)_{16} = (?)_{10}$$

2 1 0

$$\begin{aligned} 2AF &= F \times 16^0 + A \times 16^1 + 2 \times 16^2 \\ &= 15 + 10 \times 16 + 2 \times 256 \\ &= 15 + 160 + 512 \\ &= 687 \end{aligned}$$

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Binary $\longleftrightarrow$ Hex

④

$$(110101110)_2 = (\quad ? \quad)_{16}$$

8 4 2 1
0 0 1 1
 3

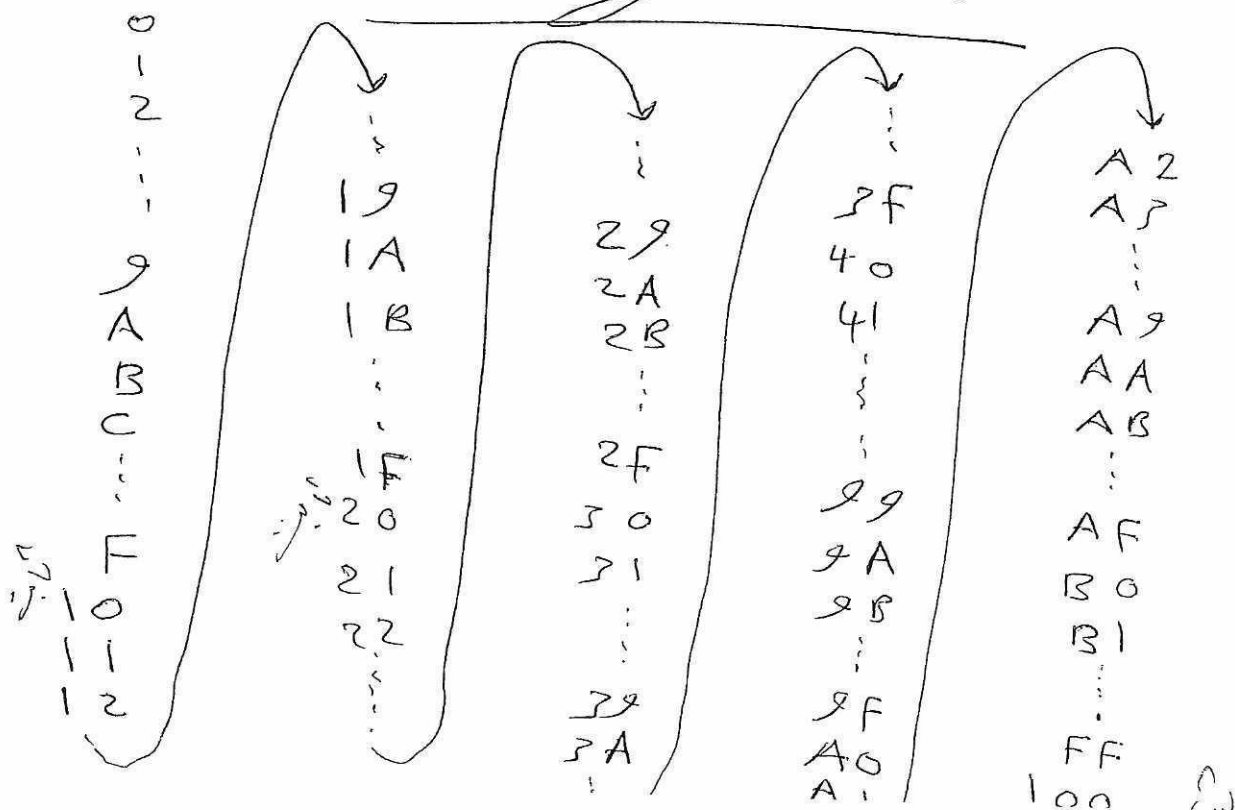
8 4 2 1
0 1 0 1
 5

8 4 2 1
1 1 1 0
 E

$$(110101110)_2 = 35E_{16}$$

$$FA_{16} = (11111010)_2$$

Counting in Hex



Binary

5

① 1's complement (first)

10100
↓ 1's
01011

② 2's Complement (second)

10100
↓ 2's
01100
بعد الواحد يتم

طريقة اخرى

10100
↓ 1's
01011
+ 1
01100
2's

قانون

2's Complement = 1's Complement + 1

Addition in binary

Ⓟ
00110
+ 00111

01101

6

$$\begin{array}{r} 01110 \\ - 00101 \\ \hline \end{array} \xrightarrow[\begin{array}{l} \text{نقص الطرف} \\ \text{الرجوع} \end{array}]{2's} \begin{array}{r} 111 \\ 01110 \\ + 11011 \\ \hline 01001 \end{array}$$

[1]
cy

$$\begin{array}{r} 01110 \text{ (2)} \\ - 00101 \\ \hline 01001 \end{array}$$

$$\begin{array}{r} 1 \text{ (16)} \\ 27 \text{ (17)} \\ - 17 \\ \hline \end{array}$$

~~Add~~ Addition in Hex

$$\begin{array}{r} 1 \text{ (1)} \\ 5C A 4 \\ + 13 E B \\ \hline 708F \end{array}$$

$$\begin{array}{r} A \quad 10 \\ E \quad 14 \\ \hline 24 \geq 16 \\ \downarrow \\ 24 - 16 = 8 \end{array}$$

$$\begin{array}{r} 1 \\ C \\ 3 \\ \hline 16 \geq 16 \\ \rightarrow 16 - 16 = 0 \end{array}$$

$$\begin{array}{c} 13 \quad 9 \quad 16 \quad 16 \\ 5 \times A4 \rightarrow 20 \end{array}$$

$$\begin{array}{r} 13EB \\ \hline 48B9 \end{array}$$

7

ASCII: American standard code for Information Interchange.

7 bit is used to code $2^7 = 128$ code (char) ^{7 bits}?

$$'A' = 100\ 0001\ B = 41h$$

$$'a' = 110\ 0001\ B = 61h$$

In new system ASCII is an 8 bit $\rightarrow 2^8 = 256$ code?

$$'0' = 011\ 0000\ B = 30h$$

$$'1' = 011\ 0001\ B = 31h$$

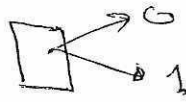
$$'2' = 011\ 0010\ B = 32h$$

⋮

$$'9' = 011\ 1001\ B = 39h$$

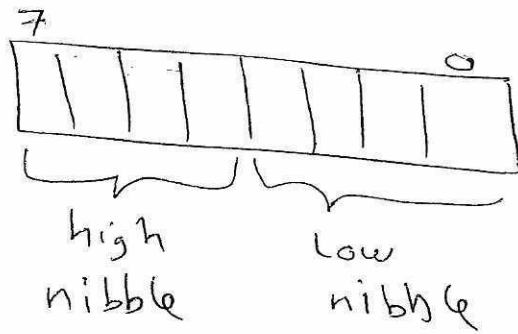
} 8 bits

* Bit (binary digit)



8

* 1 byte = 8 bit



* 1 nibble = 4 bit

1 byte = 2 nibbles

* 1 word = 16 bit = 2 byte



* 1 kilo byte (1KB) = 1024 byte = 2^{10} byte

* 1 Mega byte (1MB) = 1024 KB = 2^{20} byte

* 1 Giga byte (1GB) = 1024 MB = 2^{30} byte

* 1 Tera byte (1TB) = 1024 GB = 2^{40} byte

2^{10}

How many byte in 16 KB?

$$16 \text{ KB} = 2^4 \times 2^{10} = 2^{14} \text{ byte}$$

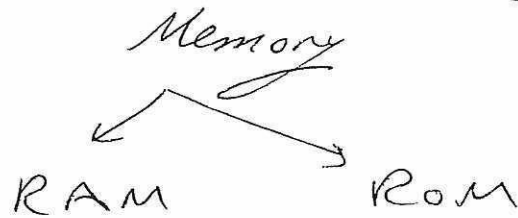
How many bit in 16 KB?

$$\begin{aligned} 16 \text{ KB} &= 2^4 \times 2^{10} \text{ byte} = 2^{14} \text{ byte} = 2^{14} \times 8 \text{ bit} \\ &= 2^{14} \times 2^3 \text{ bit} = 2^{17} \text{ bit} \end{aligned}$$

How many KB in 32 GB?

9

$$32 \text{ GB} = 2^5 \times 2^{30} \text{ byte} = 2^5 \times 2^{20} \times 2^{10} \text{ byte} \\ = 2^{25} \text{ KB}$$



[1] RAM:

- Random access memory
- used for Read and write
- It is used to store data.
- It is used for temporary storage.
- The data will lost when the power is turned off
- It is called volatile memory

[2] ROM:

- Read only memory
- used for read only.
- it is used to store data and code (program)
- It is used for permanent storage
- The data will not lost when the power turned off
- It is called non-volatile memory

computer organization

(10)

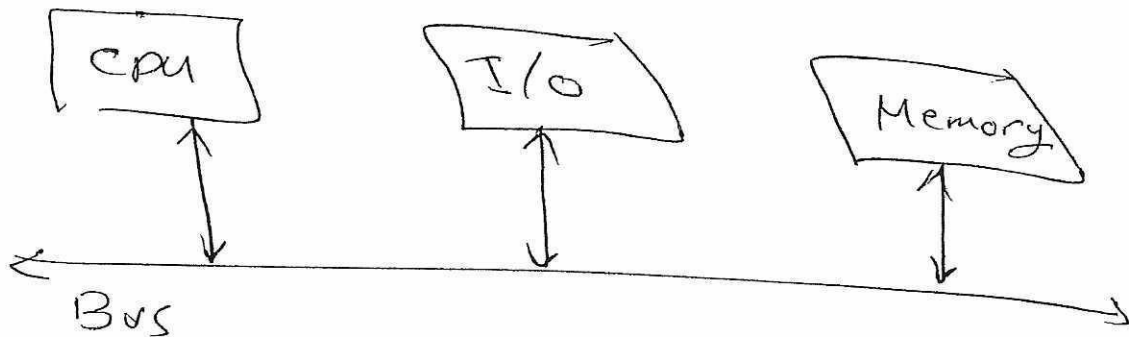
① CPU: central processing unit

executes all instructions.

② I/O devices (input/output)

توفر
provides a means to communicate with CPU.

③ Memory: store data or code



There are 3 types of bus

1 - Address bus

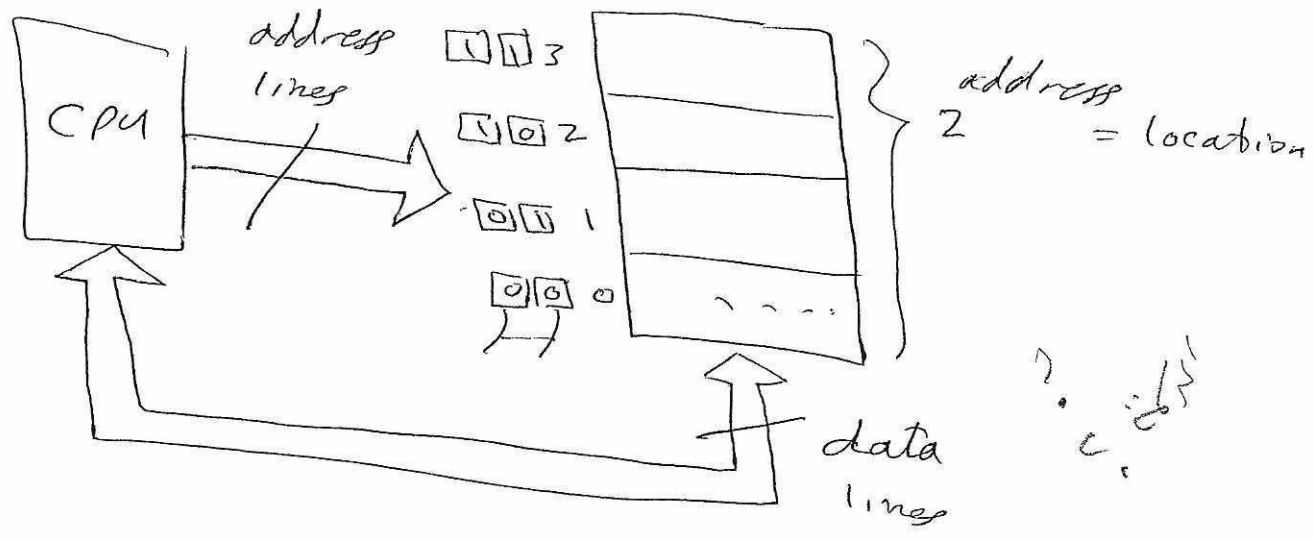
2 - data "

3 - control "

• unidirection اتجاه واحد

• bidirection اتجاهين

Address and data lines define the size of memory that can be connected with CPU.



Example: ~~what~~

organization = $2^x \times y$

x : # of address lines (pins)

y : # of data lines (pins)

capacity = $2^x \times y$ bit

Example: CPU has 16 address lines and 8 data lines. 1-organization (memory) (12)
 2-capacity ()

$x = 16$ address

$$2^{16} = 2^6 \times 2^{10} = 64K$$

$$\text{organization} = 2^{16} \times 8$$

$$= 64K \times 8$$

$$\text{capacity} = 512K \text{ bit?}$$

byte $\rightarrow$ 1 byte = 8 bits

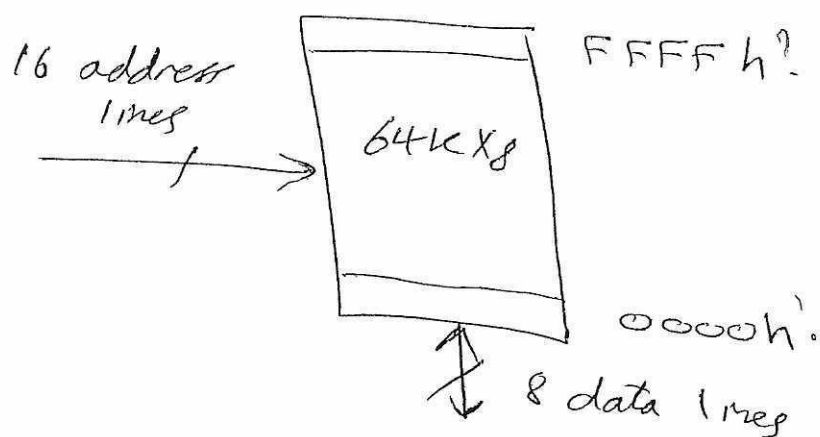
capacity in byte

$$\text{capacity} = \frac{512K}{8} \text{ B} \rightarrow \text{byte?}$$

$$= 64KB$$

- what is the range of addresses?





Example: CPBL has 14 address line
and 16 data lines

- 1- organization (memory)
- 2- capacity
- 3- capacity (in byte)
- 4- Range of addresses

$$\begin{aligned}
 \text{organization} &= 2^x \times y \\
 &= 2^{14} \times 16 \\
 &= 2^4 \times 2^{10} \times 16 \\
 &= 16 \text{ K} \times 16
 \end{aligned}$$

capacity = 256 k bit

capacity = 32 k byte

